

provide. So they have a power adapter to provide external power, thus, making them non-portable in true definition. Then again, finding a power socket next to a computer isn't that difficult a task. Laptop drives have only recently achieved the 1TB storage space limit while remaining low on power consumption. When more storage is required you'll have to go in for a 3.5 inch drive storage which can store up to 4TB of data. Also, 3.5 inch hard drives are inherently faster than 2.5 inch hard drives since they not only consume more power but also spin at a higher rate (7200 RPM) as compared to 2.5 inch drives (5400 RPM), so the transfer speeds on 3.5 inch drive are higher. However, with the recent introduction of hybrid drives that utilise flash memory along with the traditional magnetic drives have given a much needed boost to memory transfer speeds.

Flash drives

Flash drives use EEPROM (Electrically Erasable Programmable Read Only Memory) memory chips instead of magnetic disks for data storage. These are much faster than hard drives and durable too. Flash drives don't have any moving parts at all, the slide flash drives that you find on some drives aren't part of the circuitry so even if that breaks down the flash drive will work perfectly. They are also pretty shock-proof, we've had flash drives that have handled a 100 foot drop pretty good and continued to function properly. The drawback is that they are limited to a certain number of write cycles. This depends on the technology used to manufacture the drives in the first place, if SLC (Single Level Cell) memory is used then they'll handle about 100,000 write cycles and if MLC (Multi Level Cell) memory is used then expect to get about 4,000 write cycles out of your flash drive. SLC flash drives are also expensive as compared to MLC drives. When you start noticing dropping speeds while using your flash drive then you can be assured that it is nearing the end of its life and it's time to get a replacement soon. Given the high transfer speeds you can get flash drives that use either USB 2.0 or USB 3.0, the USB 3.0 drives will have a blue plastic tongue beneath the contacts or the plug.

